



**JABATAN PEMBANGUNAN KEMAHIRAN
KEMENTERIAN SUMBER MANUSIA
ARAS 7 & 8 BLOK D4, KOMPLEK D
PUSAT PENTADBIRAN KERAJAAN PERSEKUTUAN
62530 PUTRAJAYA**

KERTAS PENERANGAN (INFORMATION SHEET)

KOD DAN NAMA PROGRAM / <i>PROGRAM'S CODE AND NAME</i>	Z-009-4: 2015 CORE ABILITIES	
TAHAP / LEVEL	4 (FOUR)	
NO. DAN TAJUK KEBOLEHAN TERAS / <i>CORE ABILITY NO. AND TITLE</i>	CA05 STRATEGIC THINKING SKILL	
NO. DAN PENYATAAN KEBOLEHAN / <i>ABILITY NO. AND STATEMENT</i>	05.01 ADVOCATE EFFECTIVE THINKING 05.02 COMPLY WITH EFFECTIVE ANALYSIS 05.03 APPLY STRESS MANAGEMENT TECHNIQUES 05.04 APPLY BUDGETARY CONTROL PROCEDURES 05.05 APPLY PROBLEM SOLVING TECHNIQUES	
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TAJUK/TITLE: EFFECTIVE THINKING SKILL

TUJUAN/PURPOSE:

This information sheet is intended to describes the ability to come up with effective plans in line with an organization's objectives within a particular economic situation. The person who is competent in this CA shall be able to demonstrate, practice and apply strategic

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PENERANGAN/INFORMATION:

1. Introduction to Strategic Thinking

1.1 Thinking is an activity of mind. It works on affirmations and confirmations of comprehensions, observations, conclusions of the quests and searches of life for learning, earning and problem solving.

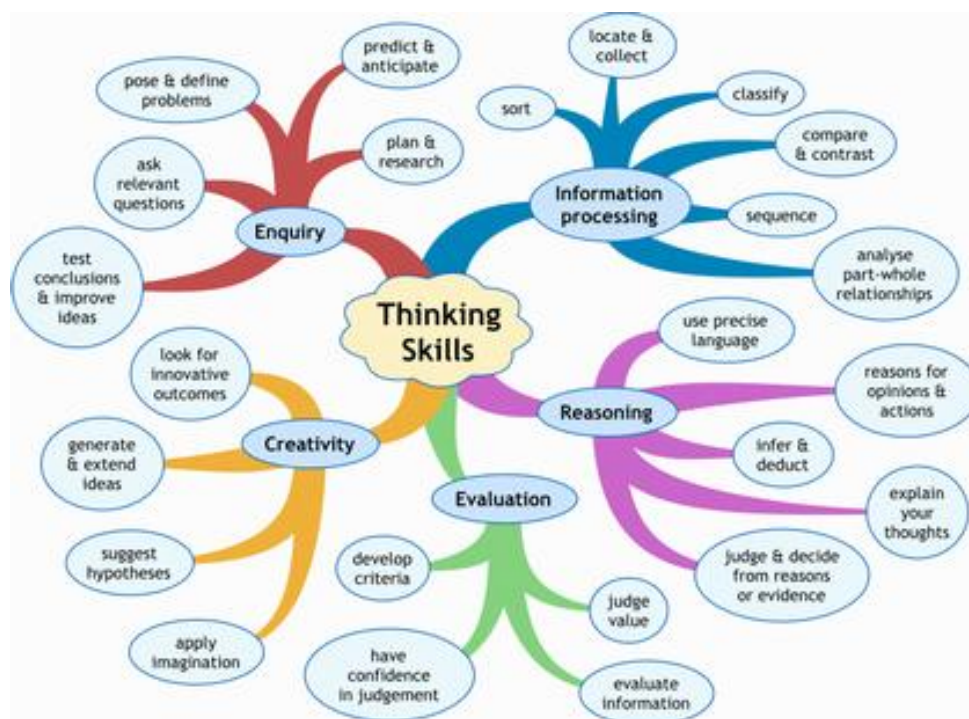


Figure 1.1: The Thinking Map

1.2 Thinking Skills are the mental processes that we apply when we seek to make sense of experience like: solve problems, make decisions, ask questions, construct plans, evaluate ideas, organize information and create objects. Thinking skills enable us to integrate each new experience into the schema that we are constructing of "how things are". There are many frameworks of thinking including Bloom's Taxonomy, DeBono's thinking tools and Lipman's modes (Moseley et al, 2005). Thinking Skills can be put into two categories; cognitive and strategic / reflective thinking skills:

i) Cognitive Thinking Skills.

Cognitive thinking skills refer to the abilities to gain meaning and knowledge from experience and information. Cognition is more than just learning information, it's

the ability to think about new information, process and speak about it and apply it to other, previously acquired information as well as developing the ability to think on higher levels, processing information more skilfully and making connections to other information more easily.

Information Gathering:	Basic Understanding:	Productive Thinking:
▪ Sensing – seeing, hearing, touching ▪ Retrieving – memory skills	▪ Organizing gathered information ▪ Forming concepts ▪ Linking ideas together	▪ Using information and understanding ▪ Creating, deciding, analyzing, evaluating

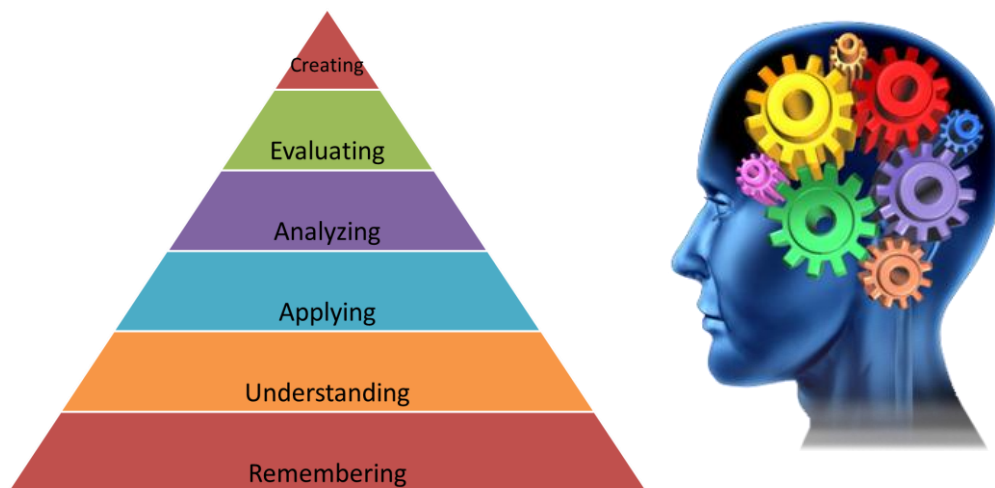


Figure 1.2(i): Domain of Cognitive Thinking Using Revised Bloom's Taxonomy

ii) Strategic/Reflective Thinking Skills:

Strategic/reflective thinking skills are metacognition - thinking about thinking. It can involve planning, monitoring and evaluating use of the cognitive skills. Reflective thinking is an active, persistent, and careful consideration of a belief or supposed form of knowledge, of the grounds that support that knowledge, and focuses on the process of making judgments about what has happened. However, reflective thinking is most important in prompting learning during complex problem-solving situations because it provides an opportunity to step back and think about how actually solve problems and how a particular set of problem solving strategies is appropriated for achieving their goal.

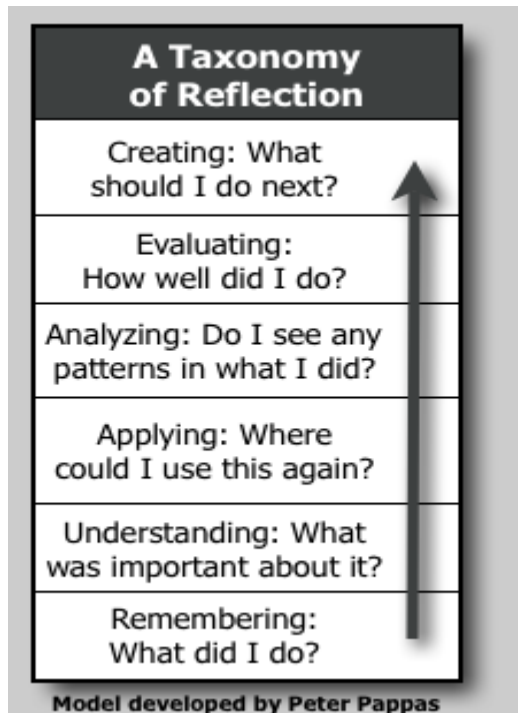


Figure 1.2(ii) : Domain of Strategic/Reflective Thinking From Bloom's Taxonomy

1.3 Effective thinking is neither **creative** thinking nor **lateral** thinking. It is being aware of and improving ones own thinking. Effective thinking moves beyond self-cantered views of the universe to a broader, more abstract realm. This means expanding thinking beyond the egocentric values and limited life experiences (Meyer, 1986). A simplified view of effective thinking is gained by identifying the major activities we carry out as we critically work through an issue:

- i) Observations. From a series of observations, we can come to establish:
- ii) Facts. From a series of facts, or from an absence of fact, we make:
- iii) Inferences. Testing the validity of our inferences, we can make:
- iv) Assumptions. From our assumptions, we form our:
- v) Opinions. Taking our opinions, we use the principles of logic to develop:
- vi) Arguments. And when we want to challenge the arguments of others, we employ:
- vii) Critical Analysis (through which we challenge the observations, facts, inferences, and so on, in the arguments that we are analyzing).

Simply, effective thinking means "using awareness + conscious thoughts + memories + reflexes, hunches, and unconscious inputs to identify current primary needs, and decide how to fill them well enough."

1.4 An Effective Thinker consciously works at making his/her thinking more rational, clear, accurate, and consistent. An effective thinker is one who utilizes a combination of critical and creative thinking within the reasoning process. Strategic Thinker who are able to assess and create opportunities for the future and implement actions that support business goals. What qualities do you need more of to become a strategic thinker? Compare these eight (8) characteristics and their opposite to determine your strategic thinking style.

STRATEGIC THINKERS	NON-STRATEGIC THINKERS
Future-based: Anticipate change and look for opportunities that may arise, or looking through the turn.	Reactive: Often wait to be told what to do or what action to take, and rarely initiate ideas.
Curious: Interested in what is going on throughout their department, company, industry and the larger business environment.	Isolated: Typically work without input from others or without understanding others' goals and objectives.
Good steward of resources: Willing to put in extra time today in order to gain a better outcome tomorrow.	Focusing on cost: Sometimes unwilling to make a good investment and do not consider the potential overall impact or long-term goals.
Risk Taker: Are not limited to past or current thinking and are willing to try new methods. They brainstorm ideas and prefer to try something new.	Cautious: Normally are only interested in maintaining the status quo.
Urgent and important: Do not equate busy with effective and place a higher value on projects with potential for a greater impact and return.	Unable to prioritize: Often treat all tasks equally without regard to impact.
Nimble: Able to adjust and modify their approach based on circumstances.	Inflexible: May be unwilling to alter their plans even when adjustments could yield a better return.
Life-long learner: Willing to be taught, proactively seeking knowledge and skill, and willing to teach others.	Satisfied: Normally not interested in learning new things or methods and are content with their current capabilities.
Creative: Consider outrageous ideas and will think outside of the box.	Predictable: Often stick with paths that are familiar.

Figure 1.4 : The Eight (8) Characteristic of Strategic/Reflective Thinker vs Conversational Thinker

1.5 There are three (3) basic principles need to be considered when applying the thinking skills:

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- i) Principle 1 - “The creation of new knowledge (to the owner) may not necessarily result in innovation but it is a precursor to the instigation of that process.” The most powerful attribute of conceptual model building is that the owner can actually create and build new knowledge in excess of that which contributed to the development of the initial model. High order thinking questions are important to encourage learners to build conceptual models of understanding. In order for learners to be able to think laterally, they must first create in their minds a conceptual framework/model of understanding. It is only then that they can build on this framework and improve upon it. If learners have only facts to work with, they cannot create new information from these alone.

- ii) Principle 2 – “Thinking is a discipline and as such calls for changes in the way we carry out our daily activities, and make decisions, and that requires a consciousness of thought.” In other words in order for learners to think about what they're doing they need to be encouraged into a discipline of questioning, and reflecting on their thoughts and thought processes and content. This is often referred to as metacognitive activity. It is the ability to think about your own thinking.

- iii) Principle 3 – “Knowledge can also be thought of as a product that can be collectively constructed/created (new knowledge).” Inevitably some educators will make the mistake of believing that unless there is concrete evidence in a physical format of the learner's understanding, then the learner does not understand the concept or principal being discussed. Lawyers are all too well aware that intellectual property (IP) is very real, and can be patented and protected, even though there is no visible product to demonstrate that intellectual capacity. Burger and Starbird supported these basic principles with five elements of effective thinking:
 - a) Understand deeply
 - b) Fail to succeed
 - c) Be your own Socrates
 - d) Look back, look forward

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e) Transform yourself.

1.6 The Underlying Thinking Principles - "Six Thinking Hats"

Human brain thinks in a number of distinct ways which can be deliberately challenged, and hence planned for use in a structured way allowing one to develop tactics for thinking about particular issues. Edward de Bono identifies six distinct directions in which the brain can be challenged. In each of these directions the brain will identify and bring into conscious thought certain aspects of issues being considered (e.g. gut instinct, pessimistic judgments, neutral facts). None of these directions are completely natural ways of thinking, but rather how some of us already represent the results of our thinking.

Each hat does not represent natural modes of thinking and must be used for a limited time only. Many will feel that using the hats is unnatural, uncomfortable or even counter productive and against their better judgment. A compelling example presented is sensitivity to "mismatch" stimuli. This is presented as a valuable survival instinct, because, in the natural world: the thing that is out of the ordinary may well be dangerous. This mode is identified as the root of negative judgment and critical thinking. Six distinct directions are identified and assigned a colour.

The six directions are:

- i) Managing (Blue) - What is the subject? What are we thinking about? What is the goal?
- ii) Information (White) - considering purely what information is available, what are the facts?
- iii) Emotions (Red) - intuitive or instinctive gut reactions or statements of emotional feeling (but not any justification)
- iv) Discernment (Black) - logic applied to identifying reasons to be cautious and conservative
- v) Optimistic response (Yellow) - logic applied to identifying benefits, seeking harmony
- vi) Creativity (Green) - statements of provocation and investigation, seeing where a thought goes.

Coloured hats are used as metaphors for each direction. Switching to a direction is symbolized by the act of putting on a coloured hat, either literally or metaphorically.

These metaphors allow for a more complete and elaborate segregation of the thinking directions. The six thinking hats indicate problems and solutions about an idea the thinker may come up with.

Edward de Bono's "Six Thinking Hats" for idea generation and evaluation

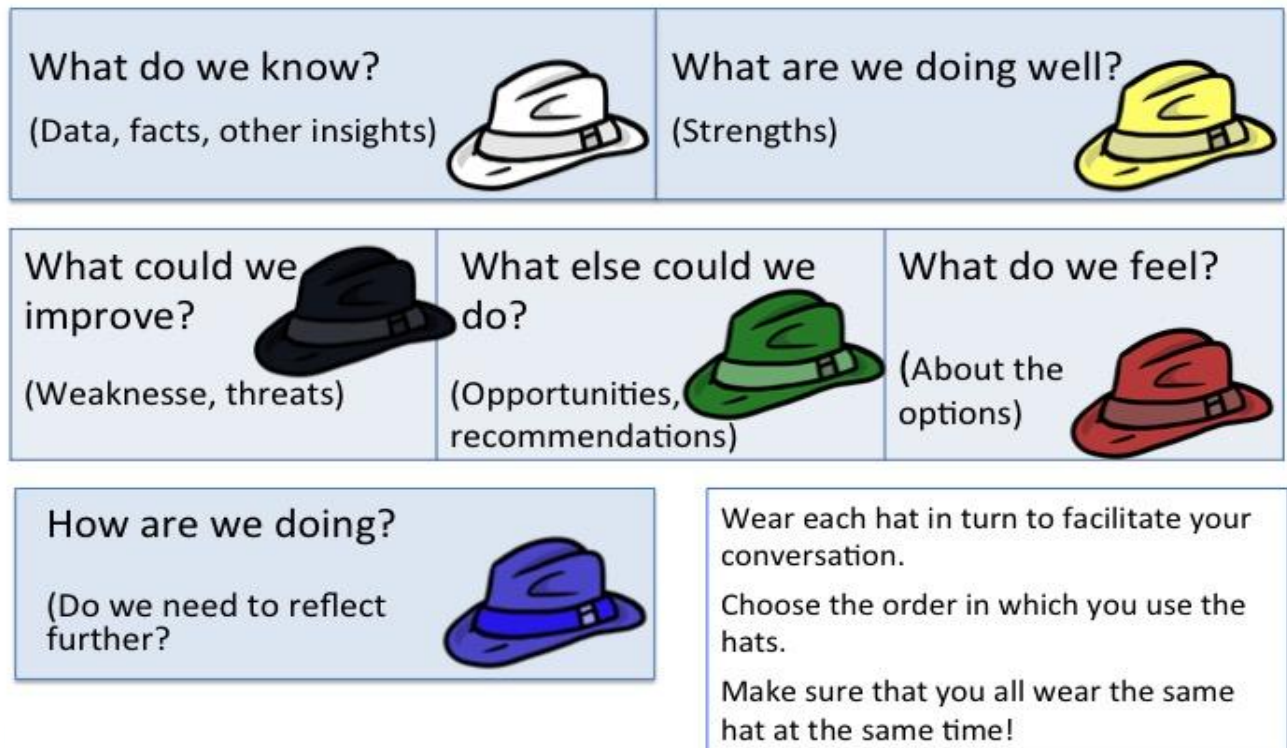


Figure 5 - Edward De Bono Six Thinking Hats for Ideas Generation and Evaluation

1.7 The Thinking Strategies and Its Application in Problem Solving

Thinking strategic is a powerful leadership tool that gives organizations the foresight and insight needed to succeed in the long term, especially when applied throughout the organization. Strategic thinking involves setting goals, developing long-range plans, anticipating the unexpected, analyzing your environment, and even cooperating with your competitors. When organizations begin to think strategically, they gain:

- i) Insight, or problem solving skills, that help them intuitively make sense of chaos in their environment.
- ii) The ability to see emerging conditions that could potentially provide long-term competitive advantage.

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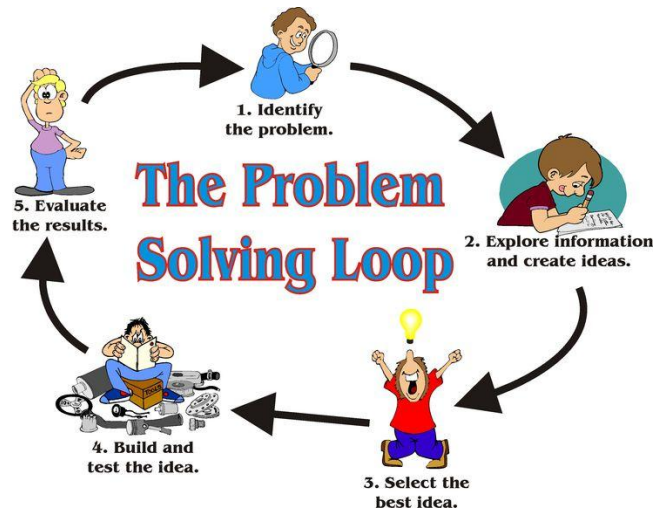
- iii) The skill of visualizing, interpreting and scanning the environment for information about the organization's present and future.
- iv) The ability to identify new market opportunities and create real solutions that advance business.
- v) The ability to understand the importance of relationship building and its interconnectedness with business goals.

What are thinking skills activities that can be used effectively in thinking strategies?

- a) Brainstorming - to promote fluent thinking
- b) Flexible Thinking - activities stretch the mind into considering possibilities beyond the usual responses.
- c) Forecasting - forecast what could be or might be.
- d) Inductive Thinking - collect, organize, and determine whether their generalizations can hold up against data.
- e) Inference Making - provide a possible consequence, conclusion, or implication from a set of facts or premises.
- f) Logical Thinking - activities that require logical thinking begin with assumptions or concepts and generate ideas step-by-step to arrive at an endpoint or a solution.
- g) Deductive Thinking - asks to consider the generalizations given and provide supporting data.
- h) Problem Solving - complex thinking process often involve problem solving and decision making.
- i) Decision Making - thinking skills needed in choosing the best response from several options.
- j) Observation - watch for a purpose and note objective changes, details, or procedures.
- k) Interpretation - use their perceptions in examining their assumptions when making a judgement or reaching a conclusion.
- l) Comparison - examine two or more situations, objects, ideas, or events and seek out relationships, similarities, or differences.

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m) Analysis - examine the problem to be solved, taking it apart, identifying its elements, and finding relationships.



SOALAN / QUESTION:

1. What do you understand with strategic thinking?
2. Explain the underlying principles of strategic thinking in solving a problem.
3. Distinguish effective thinking and strategic thinking.
4. Describe how you advocate effective thinking in strategic thinking.

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